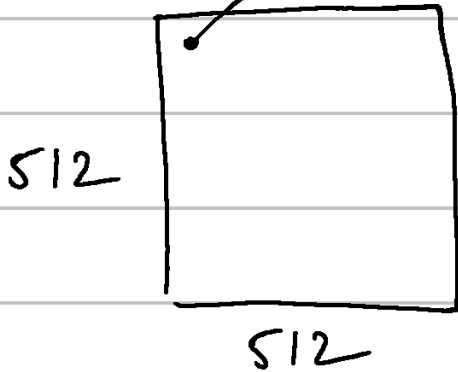


Lecture 31

Image pixel in grayscale

$$0 \leq x_i \leq 255$$



$$x \in \mathbb{R}^n$$

$$n = \underline{\underline{512 \times 512}} !!$$

image is a vector in $512 \times 512!$

Change of basis is a very useful technique in compression of signals, image, video.

JPEG is a compression for image that utilizes the fact that pixels that are nearby have similar color values.

Lets see what can be a better basis for image than standard basis $\rightarrow$

$\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} \rightarrow$ able to directly give us solid images.

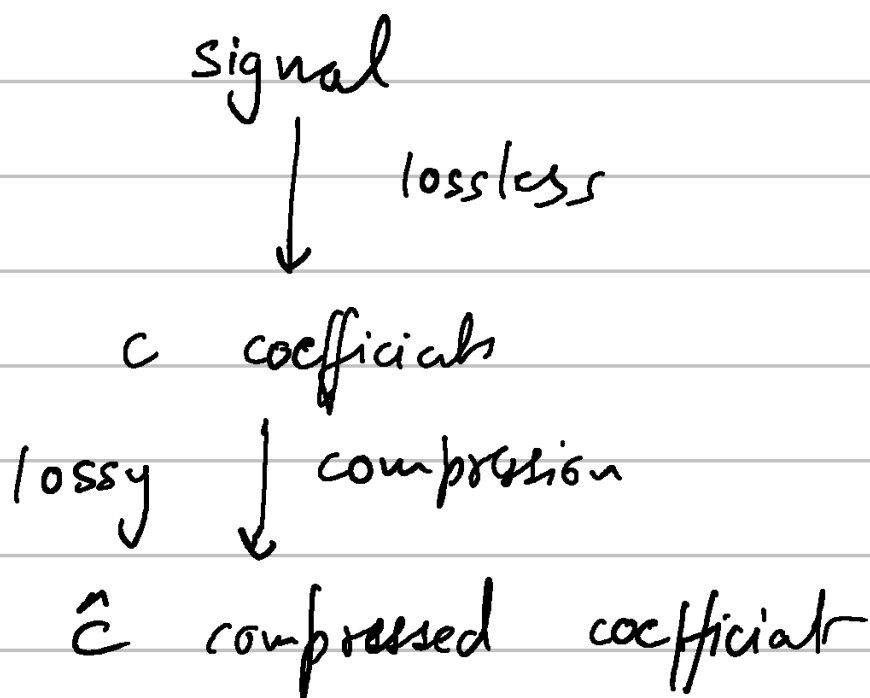
$\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \rightarrow$ able to give us contrast in close pixels.

$\rightarrow$ Depending on our data set, we can choose a basis that would help in compression of data.

JPEG uses the fourier basis (8×8)

$$\begin{bmatrix} 1 \\ \vdots \\ \vdots \\ \vdots \end{bmatrix} \quad \begin{bmatrix} 1 \\ w \\ w^2 \\ w^3 \\ \vdots \end{bmatrix} \quad \begin{bmatrix} 1 \\ w^2 \\ w^4 \\ \vdots \end{bmatrix} \quad \dots$$

→ Change of basis is a lossless step!



→ One way to compress is to just remove $\checkmark$ c's lower than a threshold.

So now let's say we are left with 3 $\hat{c}$ from 64 c

$$\hat{x} = \sum \hat{c} v_i \rightarrow \text{new basis.}$$

we have achieved compression of 64 : 3

For videos, we can use the fact that one still frame's pixels are highly correlated to the next frame.

→ Then use a correction factor to keep track of the change & compress that.

Another competition for fourier basis is the wavelets basis

$$\rightarrow \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ -1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \dots$$

for $\mathbb{R}^8$

$\rightarrow$ Lets see how to change basis and find new coefficients.

$$P = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_8 \end{bmatrix} \quad \leadsto \quad w_i = \text{Basis of wavelets}$$

$$P = c_1 w_1 + c_2 w_2 \quad \dots + c_8 w_8$$

in standard basis

$$P = \begin{bmatrix} | & | & & \\ w_1 & w_2 & \dots & \\ | & | & & \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_8 \end{bmatrix}$$

W

$$\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_8 \end{bmatrix} = W^{-1} \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_8 \end{bmatrix}$$

A good basis $\rightarrow$

- ① Fast inverse (FFT, Fast wavelet transform)
- ② Few basis vectors are enough to reproduce the signal closely.

Change of basis

We actually just did that for wavelets.

→ General case →

$$\begin{bmatrix} x \end{bmatrix}_{\text{old basis}} = \begin{bmatrix} w \end{bmatrix}_{\substack{\text{new} \\ \text{basis} \\ \text{vectors}}} \begin{bmatrix} c \end{bmatrix}_{\substack{\text{coeff} \\ \text{in new} \\ \text{basis}}}$$

$$x = Wc$$

To check

Are the coefficients dependent on the basis of x ?

It seems like we should get different values for c if it was in standard basis than Fourier.

→ T : Linear transformation

• with $v_1, v_2 \dots v_8$ it has transformation matrix A .

• with $w_1, w_2 \dots w_8$ it has transformation matrix B .

then A & B are similar.

$$\Rightarrow B = M^{-1} A M \quad \text{for some } M.$$

→ They have same eigenvalues.

What is A ??

A → matrix that does the transformation on a vector.

Suppose we have eigen vector basis. $v_1, v_2 \dots v_n$

$$T(v_i) = a_{1i} v_1 + a_{2i} v_2 \dots a_{ni} v_n$$

Now for eigenvector basis $\rightarrow$

$$T(v_i) = \lambda_i v_i$$

$A \rightarrow$ diagonal matrix with λ 's

It would be perfect to have eigenvector basis but it is computationally expensive to do so.

So we tend to find a good enough basis like wavelets / fourier.